

UV/Vis measurement of load, sludge characteristics and toxicity

Dr. Achim Gahr* and Dr. Dirk Köppenkaströp**

* STIP Isco GmbH, 64823 Groß-Umstadt, Siemenstrasse 2, Germany
(E-mail: achimg@stip.de)

** STIP Isco GmbH, 64823 Groß-Umstadt, Siemenstrasse 2, Germany
(E-mail: koeppenkaströp@stip.de)

Abstract

UV/VIS spectroscopy is a common technique for water analysis, e.g. fresh water, surface water and wastewater. However, natural limitations exist for aqueous samples of high optical density. The combination of sludge settling and UV/VIS measurement allows the analysis of wastewaters with high precision even if dissolved compounds or particulate matter are present in high concentrations. The presented technology can be applied for the quality monitoring of effluents, inlet waters and activated sludge. The evaluation of the settling behavior of the sludge is used to gain sludge quality data for process control.

Keywords

Activated sludge, sludge parameter, sludge quality, UV/Vis spectroscopy, water monitoring and control, wastewater treatment

INTRODUCTION

Effective wastewater treatment can only be fulfilled by utilizing continuous wastewater monitoring techniques and process control. Regarding ecological as well as economic aspects the efficiency and the success of wastewater treatment plants are given by different factors, such as the operational stability of the process, the quality of treated water and the potential of cost savings. In order to reach these goals stable treatment conditions are required. The measurement of organic and nutrient parameter is therefore a prerequisite to control concentrations and to calculate load.

On the other hand the quality of sludge is a basic parameter to estimate and to proof the stability of biological treatment. A continuously working monitor for sludge parameters is therefore of increasing interest. Online monitoring of sludge quality is expected to be a reliable tool to increase the efficiency of treatment processes. The cost-benefit ratio can be improved by the minimization of the sludge volume and the prevention of sludge disposal. The new developed analyzer will also allow toxicity monitoring due to the fact that toxic events diminish the quality of sludge and lead to a decrease of the sludge index as a result of upset conditions in the biomass.

This paper presents a technical solution for the product and application requirements mentioned herein. UV/VIS absorption spectroscopy has been used to develop an online and in situ probe which allows the simultaneous measurement of load and sludge parameters. The new analyzer STIP-scan will be described. In comparison to conventional cabinet and in-situ analyzers the new probe allows the measurement of nitrate, spectral absorption coefficient (SAC_{254}), chemical

oxygen demand (COD), total organic carbon (TOC), total solids (TS), sludge volume (SV) and sludge index (SI). The analyzer is almost maintenance-free and does not require chemicals for neither operation nor calibration.

TECHNOLOGY

Fundamentals

The goal of the development has been to provide a low-cost, easy to handle and powerful online and in situ monitoring probe for wastewater applications. To measure water quality parameters in the inlet, the effluent and also in the activated sludge basin UV/VIS spectroscopy has been the analytical method of choice.

The fundamentals of the technology which is presented herein are based on principle thoughts. They have been derived from both the experience of UV/VIS analysis for waste water applications and the theory of absorption spectroscopy. Compounds that cannot be seen with optic detecting devices cannot be calculated by mathematical methods. But if compounds can be directly measured by absorption spectroscopy then mathematical methods may increase the precision of the analytical result.

In order to meet the end user requirements but also to address technical limitations the advantageous of UV spectroscopy and the importance of sludge settling for the reliability of treatment processes have been combined. Utilizing synergies means the use of sludge settling for water clarification to improve the optical properties of the sample as a precondition for high precision absorption measurement. In addition, the kinetic behavior of the settling process yields to sludge parameters as the decision marks for process control.

UV/VIS spectroscopy

The new analyzer STIP-scan is a UV/VIS spectroscopic probe which operates on the principle of light absorption. The core of STIP-scan technology is a spectroscopic system existing of a light source, a sample chamber and a detector. A schematic view of the technical set-up is given in figure 1. As light source a xenon lamp is used to emit light flashes through an optical cell containing the sample. The non-absorbed portion of the light is focused by a mirror on the entrance slit of the optical fiber of a miniaturized diode array spectrometer. This is recording light intensities in the wavelength range from 190 to 720 nm. Along the optical axis two lenses are used to ensure a parallel light path in the sample cell and to collect the transmitted light.

Selected parts from the measured spectra are used for data exploitation and to calculate nitrate, the COD or TOC and the SAC_{254} as the parameters which are available by direct UV spectroscopy. For nitrate measurement a multi wavelength algorithm is used for the UV range from 200 nm to 230 nm. The spectral data between 210 nm to 400 nm are used to measure the COD. Besides the COD also the TOC can be given by correlation analysis. However, it is well known that the discrimination of different parameters only by UV absorption spectroscopy may fail due to overlapping absorption bands. As a consequence an algorithm is implemented in order

to minimize interacting absorption effects, and thus to avoid under- or overestimation. The algorithm can be described as an iterative plausibility check for nitrate and COD evaluating the overlapping spectral range from 210 nm to 230 nm. As a result the analytical precision can be increased significantly.

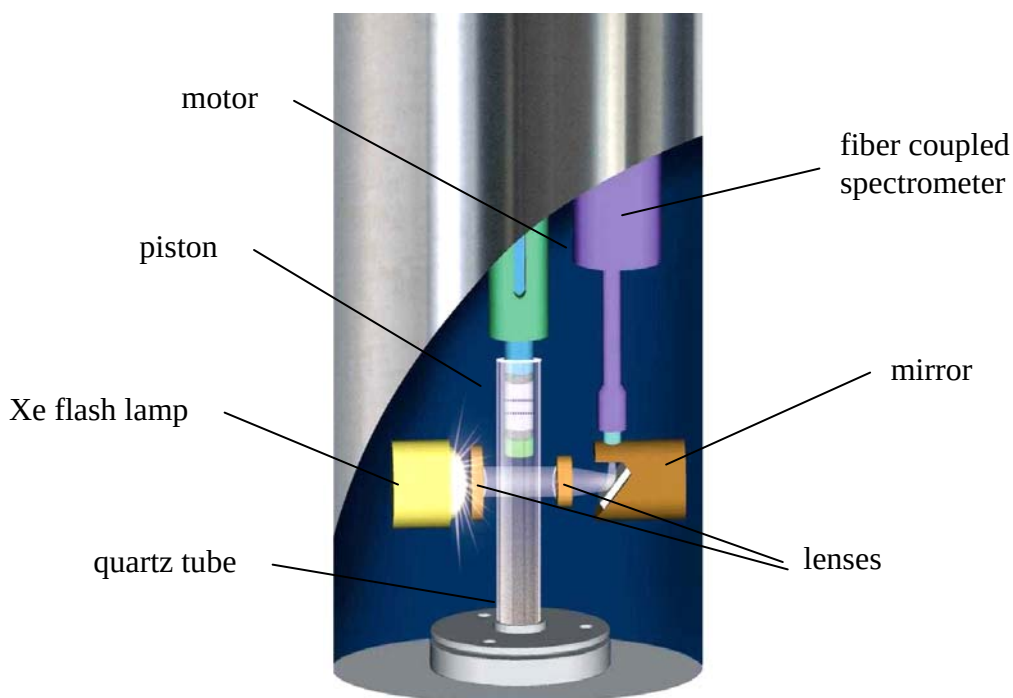


Figure 1: Schematic set-up of the UV probe showing the opto-mechanical assembly.

Sampling and settling

To realize the bifunctional design of the optomechanical part of the probe, i.e. spectroscopy along the optical axis and settling in vertical direction, a quartz tube has been chosen acting as the optical cell and the settling chamber. An optimized ratio of the tube length to inner diameter being about 80 has been chosen to achieve precise absorption data and to minimize interferences during the settling process. In order to fill the chamber the waste water is sucked by a motor driven piston into the quartz tube. The piston reaches the zero position and continuous measurements are taken with a time interval of 5 sec. The analytical wavelength for solids lies in the visible range, e.g. 600 nm. According to the time dependent behavior of the absorption (see figure 2) three regimes can be distinguished.

Regime 1:

Just after sampling the optical density of the waste water is highest due to the heterogeneous mixture of dissolved compounds and particulate matter. In this case the absorption is a more or less undefined mixture of the target analytes, such as nitrate and UV absorbing organics, and the

“background” matrix which is responsible for the limitation of direct UV measurement in waste water. In municipal waste waters it has been seen that the absorption remains constant in the first few minutes after the piston reached the starting position. This “quasi-static” behavior of the sample is characterized by an initial flocculation prior to the sludge settling.

Regime 2:

After a distinct time the dynamic process of sludge settling yields to a clarification of the waste water. This process is observed in the upper part of the optical cell, i.e. in the lightened volume of the water sample along the optical axis. Monitoring the time dependant sludge behavior is a differential kinetic approach, wherein the change of absorption is fastest and highest compared to other measuring points along the tube axis. The shape of the settling curve is a function of the sludge characteristics expressed as the decrease of UV absorption with time. The dynamic regime allows the measurement of the sludge parameters, i.e. sludge volume SV (mL/L), total solids TS (g/L) and hence the calculation of the sludge index SI (mL/g). The input data for these parameters are the slope of the settling curve, the extend of clarification given by ΔA and the settling interval Δt_s . The precision of the result can be increased by correlation analysis.

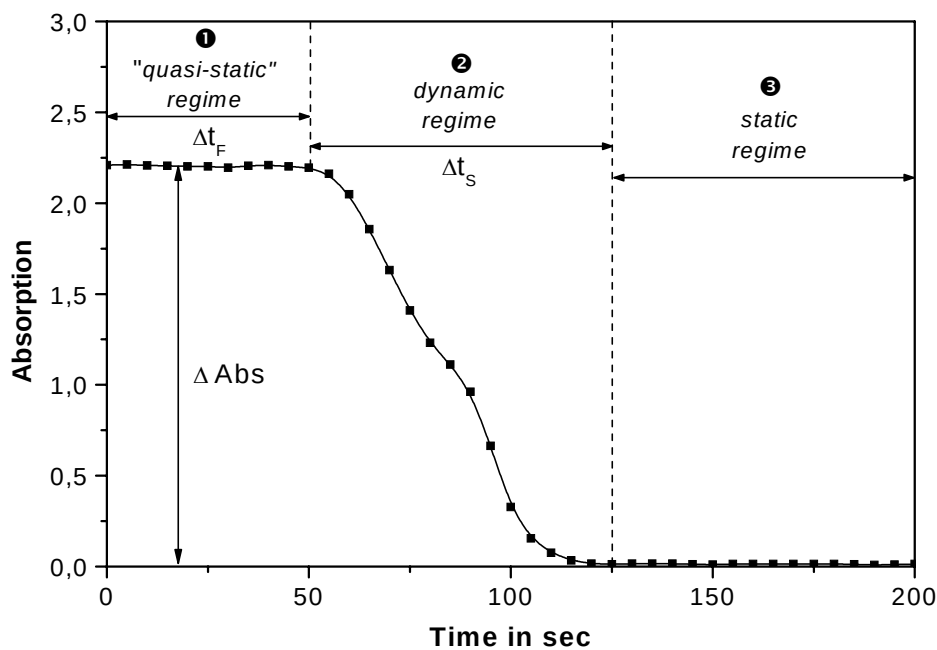


Figure 2: Kinetic curve of sludge settling in the quartz tube of the UV probe.

Regime 3:

The static regime of the settling curve describes a constant absorption of the waste water in the measuring volume of the sample which is small compared to the inner volume of the cell. Nitrate, spectral absorption coefficient (SAC_{254}) and chemical oxygen demand (COD) or total organic carbon (TOC) are measured.

Color and light absorbing inorganic salts may cause additional absorption in industrial applications. Compared to regime 1 the clarified waste water enables a very accurate and reproducible measurement of dissolved compounds. Influences caused by light scattering of particles are minimized. Depending on the settling characteristics of the wastewater, the complete measurement cycle takes between 1 and 5 minutes.

Reference measurement and interferences

Common limitations of spectroscopic approaches are the deposits on the optical windows and the decrease of light intensity of the radiation source. In order to prevent mechanical interferences the inner surface of the quartz tube is mechanically cleaned by special seals. They are mounted on the piston to remove dirt and deposits while filling and emptying the sample chamber.

However, it is well known that coloration of glass and the deposit of salts or other thin light absorbing films may cause persistent interferences which cannot be removed by mechanical wiping systems. The advantageous design of the UV probe offers a simple and powerful possibility to measure a reference absorption. The piston contains a bore hole which is used to measure a zero signal through air. Any changes in the optical properties of the windows as well as the drift of the light source can then be compensated automatically.

EXEMPLARY RESULTS

Common applications and future trends

The STIP-scan probe has been developed for municipal and industrial wastewater treatment plants for different applications. An overview of applications and the correlating parameters is given in table 1. The multi parameter analyzer can be installed in the inlet and outlet and also in the aeration basin to control and regulate the activated sludge treatment. Depending on the monitoring site the target analytes are load parameters, i.e. COD and TOC, nitrate and sludge parameters. Also the SAC at 254 nm, a well established parameter to indicate organic loads in natural waters, can be measured. Apart from wastewater treatment plants river monitoring applications and the measurement in sewer systems are of great interest.

Particular applications in the industrial field for process control can utilize the whole absorption spectra as an additional source of information. Specific organic compounds often exhibit a characteristic absorption behavior. Distinct absorption values can therefore be used as an alarm level to recognize malfunctions of production processes and to detect environmental hazardous concentrations of key compounds.

One of the most important conditions to ensure operational stability and efficiency of a treatment plant is a high quality of sludge in the aeration tank over time. The continuous and isochronic measurement of sludge and load parameters is a tremendous advantage for real-time monitoring.

This is due to the high potential of saving running costs on the short term. On the long term recording sludge data will be used to identify crawling toxicity effects. Indeed it is an overall observance that either the quality of sludge decreases slowly with time or special efforts are necessary to ensure stable sludge properties. Ubiquitous environmental hazardous compounds such hormone like or biological active substances probably cause those effects. The impact of exceeding threshold values on the sludge index can be measured continuously with STIP-scan.

Table 1: Applications for the STIP-scan monitor.

		Load Parameter	Nutrient Parameter	Sludge Parameter	Others
River monitoring		COD, TOC, SAC	NO ₃ ⁻		
Sewer		COD, TOC, SAC	NO ₃ ⁻	TS	
Municipal WWTP	inlet	COD, TOC, SAC			
	aeration basin	COD, TOC, SAC	NO ₃ ⁻	TS, SV, SI	
	outlet	COD, TOC, SAC	NO ₃ ⁻		
Industrial WWTP	inlet	COD, TOC, SAC			spectra
	aeration basin	COD, TOC, SAC	NO ₃ ⁻	TS, SV, SI	spectra
	outlet	COD, TOC, SAC	NO ₃ ⁻		

Municipal and industrial field experience

During product development the STIP-scan analyzer has been tested with waters from various treatment plants. The objectives of the testing procedures have been the optimization of data exploitation to verify the instrument under real wastewater conditions. For reference analysis wastewater samples have been collected and were measured by standard lab methods. Some of the results are explained in the following figures.

A first correlation of nitrate measurement has been performed with a final effluent of a municipal wastewater treatment plant (figure 3). The data has been gathered this year from July 16th to August 27th. A similar correlation has been performed for the COD. The data which are summarized in figure 4 were obtained for the inlet and final effluent at the same municipal treatment plant. In both cases a linear relationship can be derived that allows the measurement of nitrate and COD in treated water. However, for the inlet a stronger variation of the COD measures has been observed compared to the final effluent values.

In order to improve the precision of measuring absorption and calculation concentrations the algorithm for data exploitation has been optimized continuously, e.g. a simple onsite calibration step has been included to adapt the instrument to different applications. As a result figure 5 summarizes two daily traces for nitrate which were measured at the same monitoring site as described above (figure 3). A very high precision of the STIP-scan analyzer can therefore be concluded for the determination of nitrate in municipal effluents.

The feasibility of the analyzer for industrial applications could be shown by analyzing samples from treatment plants for chemical wastewaters. Inlet and effluent samples have been taken to measure the nitrate concentration by online instrumentation and lab method. From the results in figure 6 and 7 it can be seen that even low nitrate concentrations can be measured in waters which contain COD from compounds used for chemical production of polymers. Even a 100fold excess of COD compared to nitrate did not influence the determination of nitrate in the inlet of the treatment plant. This result confirms that the algorithm is a reliable tool for an accurate nitrate measurement by the minimization of overlapping absorption bands.

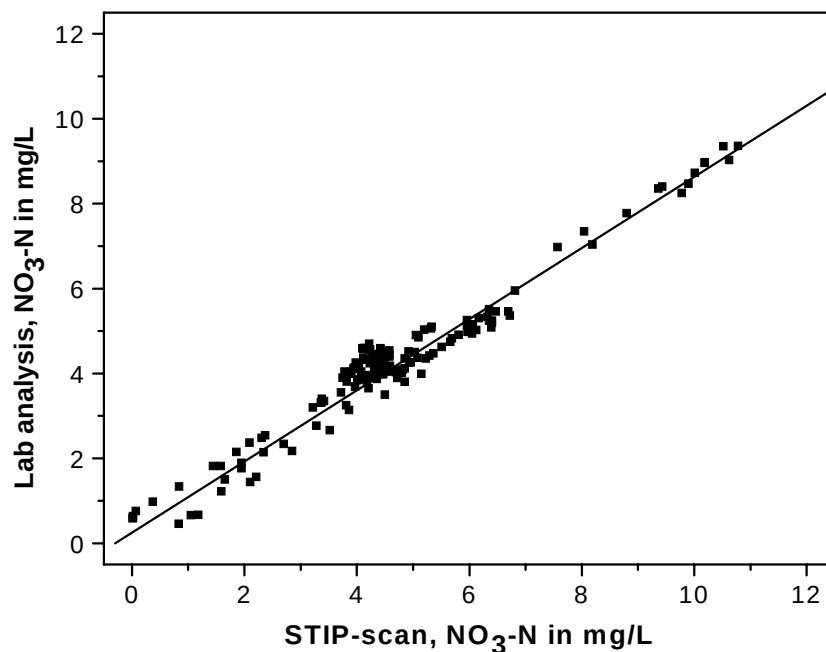


Figure 3: Measurement of nitrate in the final effluent of a municipal wastewater treatment plant (data from 16.07.02 – 27.08.02, Reinheim, Germany)

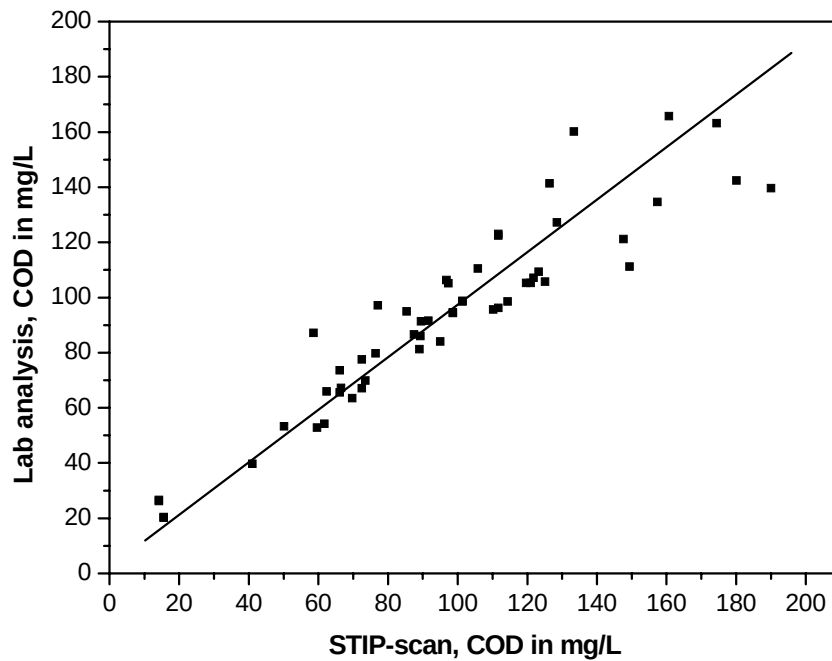


Figure 4: Measurement of COD in the inlet and final effluent of a municipal wastewater treatment plant. (data from 02.07.02 – 09.07.02, Reinheim, Germany)

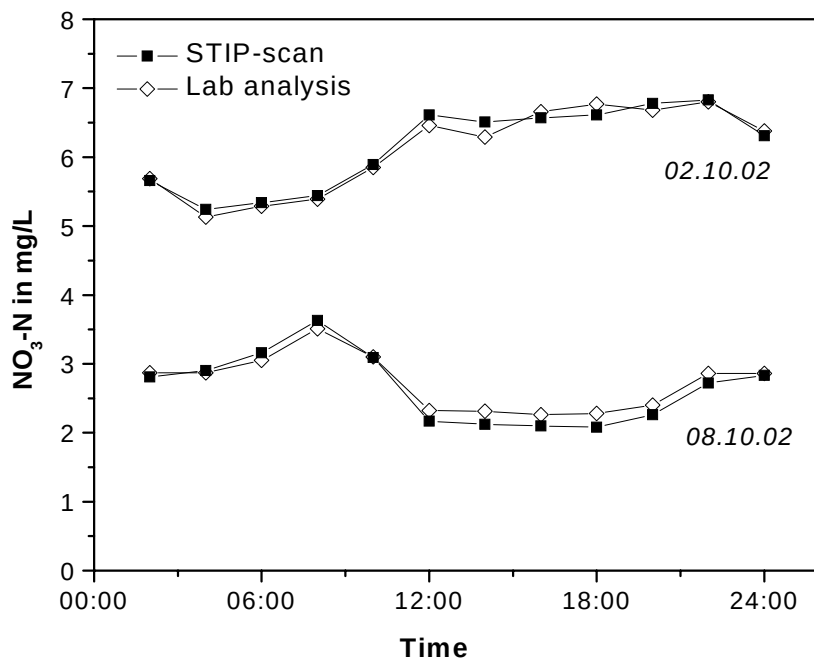


Figure 5: Measurement of daily traces of nitrate in the final effluent of a municipal wastewater treatment plant (Reinheim, Germany).

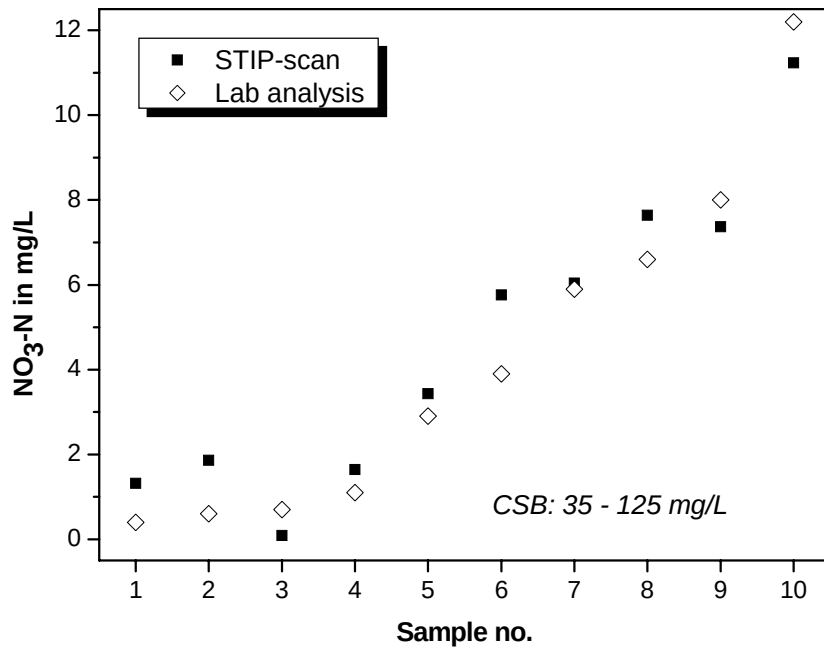


Figure 6: Measurement of nitrate in the final effluent of an industrial wastewater treatment plant (Kelsterbach, Germany).

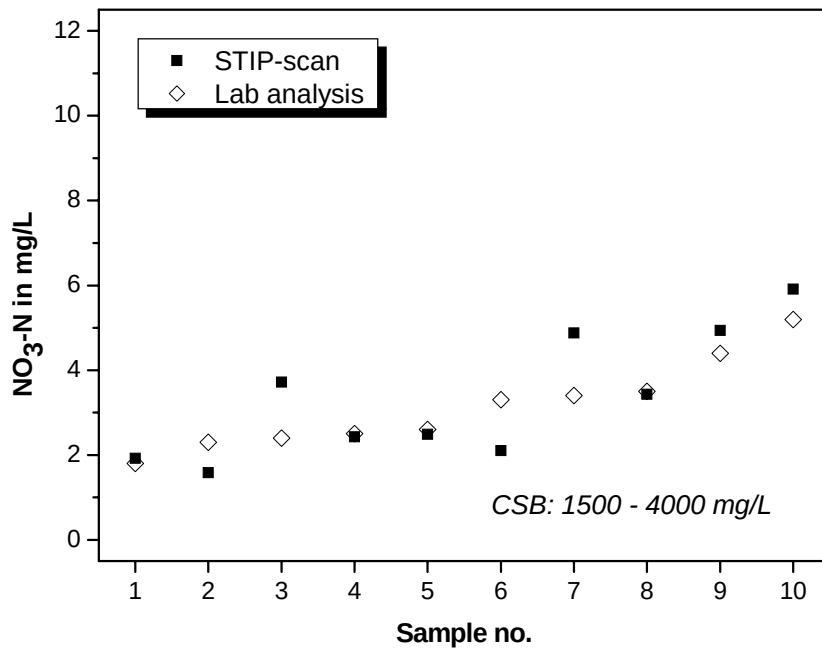


Figure 7: Measurement of nitrate in the inlet of an industrial wastewater treatment plant (Kelsterbach, Germany).

CONCLUSIONS

The development of the STIP-scan analyzer provides a simple and accurate in situ and online tool to monitor nitrate, load parameters and also the quality of sludge. Stable treatment conditions and the avoidance of sludge are the primary goals that can be reached with STIP-scan technology. Additionally, specific absorption of water constituents from industrial sites can be measured and used for process control. Besides the variety of applications the customer benefits can be concluded as follows:

- up to six (and more) parameters can be measured at the same time with one single analyzer
- in situ measurement without sample preparation
- easy to handle and user friendly
- maintenance free
- low operational costs
- high precision and accuracy of data
- no chemicals are needed

This analyzer is a low cost, low effort approach to provide accurate and precise analytical data for process control in municipal and industrial wastewater treatment plants.